Exploring an Artificial Intelligence Model Based on Integer Linear Programming for Allocating Students to Industry Placements

Moumena Salah Yassen[1,3], Victor Sanchez-Anguix[2], Juan M. Alberola[3(✉)], Fulgencia Villa[2], Vicente Julian[3], and Vicente Botti[3]

[1] College of Computer Science and Information Technology, Kirkuk University, Kirkuk, Iraq
moumena.salah@uokirkuk.edu.iq

[2] Instituto Tecnológico de Informática, Universitat Politècnica de València, València, Spain
{vsanchez,mfuvilju}@eio.upv.es

[3] Valencian Research Institute for Artificial Intelligence, Universitat Politècnica de València, València, Spain
{jalberola,vinglada,vbotti}@dsic.upv.es

Abstract. Allocating students to industry placements is a necessary task in certain educational contexts, both in Vocational Educational Training and at the University level. This refers to the process of assigning students to specific work placements within companies, where they can gain practical experience related to their field of study. In such scenarios, preferences may play a crucial role in the allocation process. This article proposes a multi-objective integer linear programming model for allocating students to industry placements. The proposal takes into consideration competence vectors, which allow the expression of aspects such as the current competences that a student has developed, the competences a student wishes to develop in a placement, or the competences required by a company for a specific placement. In this paper we study the scalability of the proposed model and what factors may affect the scalability and time performance of the model.

Keywords: Artificial intelligence · education · optimization

1 Introduction

In educational contexts, students undertake placements in companies during their training. This type of training is usually seen as beneficial for both companies and students, as companies do not incur significant costs, and students can gain valuable professional experience for their future [3]. In this regard, there is fairly widespread consensus on the positive effects of these internships, such as improved job prospects, better salaries, or higher satisfaction [8]. Traditionally,

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2024
C. Herodotou et al. (Eds.): MIS4TEL 2024, LNNS 1171, pp. 90–102, 2024.
https://doi.org/10.1007/978-3-031-73538-7_9

educational institutions manually assign students to industry placements, which may be time-consuming and cognitively complex for decision-makers. We refer to the problem of assigning students to industry placements or internships as the student-industry placement allocation (SIPA) problem.

The proposal presented in this paper extends preliminary work [10] by evaluating the scalability and performance of a multi-objective integer linear programming (ILP) model for the SIPA problem. The model takes into account different types of preferences associated with students and companies. Unlike other preference models used in similar allocation problems in education, we model preferences as three competence vectors: the current competences that a student has developed, the competences the student would like to develop in a placement, and the competences that are required by a company in a specific placement. Additionally, in this model, we incorporate other factors that are also relevant, such as the distance to the companies and the interdependence of placements. The main objective and contribution of this paper is to understand how the model scales under different scenarios and to understand what factors may affect the most the time performance of the model. The rest of the paper is organized as follows. Section 2 presents related work on assignment problems in educational contexts, with a special focus on works addressing SIPA problems. In Sect. 3, we detail the integer linear programming model under study. Section 4 describes various experiments to validate our proposal. Finally, in Sect. 5, we present some concluding remarks and outline future lines of work.

2 Related Work

In the literature, we can find proposals for assignment problems in educational contexts. For instance, partitioning students into teams [12], assigning a student(s) to a project, as well as assigning a lecturer to supervise a student's project. Although these problems are different from SIPA problems, they share common actors such as students, projects, or lecturers.

One of the problems that has generated significant interest in the scientific community is the student-project allocation problem (SPA), where the goal is to optimally assign students to projects. Typically, these problems use to define students' preferences for various offered projects, with the optimization objective being to maximize student satisfaction based on their preferences. For example, in [5] a genetic algorithm is presented to solve this allocation problem by considering the preferences of students for the projects. Other works introduce additional complexity by involving lecturers who will supervise student mentoring. In these cases, some works such as [6], also introduce preference rankings associated with each available lecturer for mentoring students. The work [9] considers departmental preferences when accepting students, while [2] takes into account the professors' capacity to supervise projects, and [15] aims to balance the workload among different departments. Another variant of these problems considers students' preferences for being mentored by a specific lecturer and the capacity (number of students) of the lecturers [11]. A similar problem is addressed in [13],

where a genetic algorithm is proposed while considering a range of supervision capacities between a minimum and maximum, in addition to student preferences. The proposal of [14] also integrates lecturer preferences for mentoring students.

Other proposals address the group-project allocation problem (GPA). Optimization in these cases involves assigning a group of students to a specific project. Proposals like [4] consider student preferences based on project topics and propose group assignments of up to three students based on these preferences. Moreover, incorporating a group of students in the allocation process leads to considering other requirements. In this regard, works like [7] take into account the skills needed for a project and the skills that a student wants to develop while working on that project, which can be related to preferences.

The proposal addressed in this paper is focused on the SIPA problem. In this type of problem, students are assigned to industry placements, that are offered by different companies. Related to this, in [16] the authors tackle the issue of allocating students to industry placements. In the model proposed in this work, each company can offer one or more placements, and companies interview students to define their preferences. The model aims to optimize company satisfaction but does not take into account student preferences. The work [1] presents a greedy algorithm for assigning students to student placements. In this case, real data is used to assign 127 students to placements offered by companies located in 23 different towns. These placements can be supervised by tutors from the students' university and lecturers from other universities closer to the companies. This work also does not take into account aspects related to preferences. Instead, the optimization problem focuses on minimizing the distance between assigned professors and the internship locations.

While many of the above works have successfully addressed various aspects of student allocation problems, they present some limitations. Preferences have typically been addressed as indications of a student's interest in one or more specific projects or a lecturer's interest in supervising one or more specific students. However, this way of modeling preferences requires defining direct evaluations that can introduce biases, especially when there are a large number of possible assignments. In this regard, our model aims to advance the state of the art in SIPA problems by incorporating preferences through competence vectors. Specifically, we consider the current level of competence development of a student, the competences that a student would like to develop during a company placement, and the competences demanded by a company for a particular placement. Hence, our method aims to optimise both student and company satisfaction, providing a more balanced and realistic solution to the SIPA problem.

3 Integer Linear Programming Model

We start by formalizing the SIPA problem. Let us define the set of students to be allocated by an educational institution as $\mathcal{S} = \{1, \ldots, s\}$. Let $\mathcal{P} = \{1, \ldots, p\}$ denote the placements that the education center has available for students. Please, note that, in this particular model, it is assumed that $p \geq s$ so that

it is possible to assign every single student to an industry placement. This is a common setting in Spanish Vocational Educational Training (VET) programs, where industry placements are compulsory for final-year students. Therefore, the educational institution should offer at least one possibility to each student. In addition to this, $\mathcal{B} = \{1, \ldots, b\}$ represents the set of companies that offer any placement for the educational center. Each placement $j \in \mathcal{P}$ is associated with a given company $k \in \mathcal{B}$. Without losing any generality let us define by $P_k \subset \mathcal{P}$ as the set of placements that are offered by company k. Each company $k \in \mathcal{B}$ has agreed with the educational center a minimum number of students to be allocated q_k and a maximum number of students to take as placement students, Q_k. In practice, some companies offer placements that have interdependencies between them. This is typically the case when these placement positions are bound to be working on different parts of the same project. Thus, if any of these placement positions are not covered, the company's project would not be completed. Let us define $\mathcal{I} = \{1, \ldots, n\}$, with each $d \in \mathcal{I}$ being a set of placement positions that have interdependence. That is, $d \subset \mathcal{P}$.

From the perspective of an academic institution, the problem of allocating students to placements consists of successfully allocating a placement for each student, considering both the preferences of students and the preferences of companies, given the requirements formulated in the previous paragraphs. Following, we show the details of the multi-objective integer linear programming model for tackling this problem based on competences. This multi-objective integer linear programming model is aimed to be used by a single academic institution or a centralized institution to match students to available placements. The main features of this model are: (i) its ability to consider both the students' and the companies' preferences using a multi-objective approach; (ii) the use of competences as a key element when matching students to placements; (iii) the inclusion of distance as a factor for considering the utility reported by an industry placement to a student; (iv) and the ability to model interdependencies between placements.

We employ $\mathcal{C} = \{1, \ldots, c\}$ to refer to the set of competences that are relevant for the assignment of students to placements. Competences are important as they provide a common ontology for businesses and students to describe both the capabilities needed by an industry placement and the students' preferences to develop or use certain capabilities. In addition to this, education and assessment based on competences is nowadays a core part of the European Higher Education Area. Hence, education institutions can also provide a factual and multifaceted evaluation of the level of acquisition of competences by students, thus acting as an objective evaluator. These competences could include course-specific competences (e.g., ability to develop network programs in C++, ability to design basic linear programming models, etc.) and also generic competences related to teamwork, oral communication, etc.

The proposed model assumes that each placement $j \in \mathcal{P}$ has associated a vector of competence relevance $\Phi_j =< \phi_{j1}, \ldots, \phi_{jc} >$, where each ϕ_{jl} represents the importance of competence of $l \in \mathcal{C}$ for the development of placement j. Similarly, a student i has two vectors associated with competences. $\Psi_i =< \psi_{i1}, \ldots, \psi_{ic} >$, where each ψ_{il} represents the current development of competence $l \in \mathcal{C}$ by stu-

dent i, and $\Theta_i =< \theta_{i1}, \ldots, \theta_{ic} >$, where each θ_{il} represents the preference of student i for developing competence l during her placement. For the sake of simplicity, it is assumed that $\sum \phi_{jl} = 1$ with $0 \leq \phi_{jl} \leq 1$, and $0 \leq \theta_{ir}, \psi_{ir} \leq 1$. It should be highlighted that, in practice, we expect the number of relevant competences to be general and limited. Thus, these vectors can be easily elicited from stakeholders. For instance, Φ_j can be elicited from the company that offers the placement when agreeing placements with schools, Ψ_i can be elicited from the academic institutions, and Θ_i can be elicited from a simple questionnaire given to students before being allocated their placements.

We formally define the utility reported by placement j to student i as $\tau_{ij} = w_{id} \times (1 - d_{ij}) + w_{ic} \times (\Phi_j \cdot \Theta_i)$ where w_{id} is the importance given by student i to the distance from her home to the allocated placement, w_{ic} is the importance given by the student to the competences to be developed in the placement, and d_{ij} is the distance between the student's home and the workplace of the placement. This distance is normalized between 0 and 1 with a min-max scaler for each student. Distance is included as a factor to be considered since it was highlighted by consulted experts as a major element in allocating students to placements. The main rationale is that many students have a young age and many do not have a driving license. Therefore, very distant placements may be unreachable by students or result in long commuting times using public transportation. The formula also includes the dot product of Φ_j and Θ_j as an average of the preferences of students with respect to competences weighted by the importance of each competence for the placement. The utility reported by student i to placement j, is defined as $\eta_{ij} = \Phi_j \cdot \Psi_i$ where the formula represents an average of the student's competences weighted by the importance of each competence for the placement. Given this, we formalize a multi-objective ILP model.

$$\max \quad \frac{\alpha}{|\mathcal{S}|} \sum_{i=1}^{s} \sum_{j=1}^{p} \tau_{ij} \times x_{ij} + \frac{\beta}{|\mathcal{S}|} \sum_{i=1}^{s} \sum_{j=1}^{p} \eta_{ij} \times x_{ij} \tag{1a}$$

s.t.

$$\sum_{j=1}^{p} x_{ij} = 1 \qquad \forall i \in \mathcal{S}, \tag{1b}$$

$$\sum_{i=1}^{s} x_{ij} \leq 1 \qquad \forall j \in \mathcal{P}, \tag{1c}$$

$$\sum_{i=1}^{s} \sum_{j \in P_k} x_{ij} \leq Q_k \qquad \forall k \in \mathcal{B}, \tag{1d}$$

$$\sum_{i=1}^{s} \sum_{j \in P_k} x_{ij} \geq q_k \qquad \forall k \in \mathcal{B}, \tag{1e}$$

$$\left(\sum_{i=1}^{s} \sum_{j \in I} x_{ij} \right) - |I| \times \delta_I = 0 \qquad \forall I \in \mathcal{I} \tag{1f}$$

where $x_{ij} \in \{0,1\}$ are binary decision variables that represent if student i is allocated to placement j, $\delta_I \in \{0,1\}$ are binary decision variables that indicate if interdependent placements $I \in \mathcal{I}$ are allocated or not. The multi-objective aspect of the problem is modeled by means of a weighted sum of the two objectives. Each objective is represented by one of the two terms in the function. The first term represents the average utility of allocations from the students' perspective, and it is related to the overall satisfaction of students with the allocation. The second term represents the average utility reported to businesses by the proposed allocations, and it is related to the overall satisfaction of companies with the allocation. These two terms are weighted by α and β which represent the importance given by the education institution to the students' and companies' satisfaction respectively, with $\alpha + \beta = 1$.

In addition to this, constraints 1b guarantee that each student is allocated to exactly one placement, constraints 1c ensure that each placement is allocated at most once, and constraints 1d and 1e guarantee that each company is allocated a number of students within the minimum and maximum agreed number of students. Constraint 1f is designed to force placements with interdependencies to be allocated all together in case that they are allocated.

As the reader may observe, the model also allows us to tackle the problem from a single objective perspective. By setting $\alpha = 1$ and $\beta = 0$ one can solve the problem by just considering the objective associated with the students' satisfaction. Conversely, by setting $\alpha = 0$ and $\beta = 1$ we can solve the problem by just considering the objective associated with the companies' satisfaction. Both objectives are measured on a 0–1 scale. Therefore, the weights α and β are related to the importance given by the educational institution to each objective. Depending on its criteria, they can set both accordingly.

4 Experiments

As suggested, one of the objectives of this article is to analyze the scalability of the proposed model and gain an understanding of what factors may affect the time performance of the model. To make this analysis, it is necessary to generate varied and different scenarios and test the proposed model under those different conditions. We start by describing how different problem instances with different characteristics were generated. Then, we describe the experiments that have been carried out and analyze the results to understand the scalability of the model and the factors that affect its scalability.

4.1 Instance Generation

We created several problem instances by changing the type and distribution of the different inputs of our linear integer programming model. Next, we specify what characteristics were taken into consideration when generating instances.

– **Number of students:** The number of students plays a major role in the size of the problem. The more students one has to allocate, the more time it

will take. In order to study the scalability of the model with the number of students, we generated problem instances with 100, 200, 300, and 400.

- **Number of placements:** Another factor that may affect the problem size is the number of placements available. The more placements, the more decision variables will be necessary, as the number of decision variables is exactly equal to $|\mathcal{S}| \times |\mathcal{P}|$. Any problem with $|\mathcal{S}| > |\mathcal{P}|$ is non-feasible. We will use the term surplus to refer to the difference between the number of placements and the number of students $|\mathcal{P}| - |\mathcal{S}|$. For each number of students, we generated problem instances that have a 0%, 10% and 20% of surplus.
- **Students' homes and companies' locations:** Students' homes are randomly sampled from points in 21 small cities in the metropolitan area of Valencia (Spain), and Valencia. Companies' locations are randomly sampled from 200 real company locations in Valencia and surrounding areas.
- **Student competence development Ψ_i:** We aim to create instances that are distinct from each other, regardless of the institution. Therefore, we have chosen to consider six competencies for these experiments. We generated students with a high, intermediate and low competence development. To match realistic proportions, when generating a student its competence development is assigned to high, medium and low competence development with 5%, 45% and 50% probability. For students with a high development, two random competences follow a normal distribution with $N(\mu = 0.8, \sigma = 0.2)$, while the rest follow a normal distribution with $N(\mu = 0.4, sigma = 0.4)$. In the case of students with a medium development one competence follows a normal distribution with $N(\mu = 0.8, \sigma = 0.2)$ and the rest follow a normal distribution with $N(\mu = 0.2, \sigma = 0.4)$. For students with a low development, all the competences follow a normal distribution with $N(\mu = 0, \sigma = 0.1)$. All of random numbers generated were truncated to the [0,1] interval. Thus, students with a high competence development will tend to have 2 strong competences, students with an intermediate competence development will tend to have one strong competence, and students with a low competence development will tend to have no strong competence.
- **Student competence preference Θ_i:** The preferences of each student with respect to the development of competences during the placement is generated in a way that one random θ_{il} has a value of 1 (i.e., the most preferred competence), one has a value of θ_{il} of 0.5 (i.e., the second most preferred competence, and the rest are generated from a uniform distribution $U(0, 0.3)$, being options with a low preference for students.
- **Student preferences for distance** In this case, the preference of student i with respect to distance w_{id} is generated from a uniform distribution $U(0, 1)$ and w_{ic} is calculated as the complement $1 - w_{id}$.
- **Placement competence importance Φ_j:** We understand that different placement options may require of one important or several important competences. We generated some scenarios where l random competences are the primary ones for the placement, and another competence acts as a secondary competence for the placement. We refer to these scenarios as l primary competence scenarios. In these scenarios, Φ_j is generated with l random competences

taking $\phi_{jl} = 1$ (i.e., the primary competence), and a random competence taking $\phi_{jl} = 0.5$ (i.e., the secondary competence), and the rest being uniformly generated from $U(0, 0.3)$. We generated scenarios with one, two and three primary competences.
- **Placement interdependence:** A company with at least 2 placements may state interdependence between some of their placements (30% probability).

For each possible configuration we generated 5 different problem instances. Thus, in total we have generated $4 \times 3 \times 2 \times 5 = 120$ problem instances. These instances will be employed to study the performance and scalability of the model.

4.2 Experimental Design

As any integer linear programming model, it can be solved via a variety of solvers. In this case we are interested on studying how the solver scalers both with commercial and free solvers. While the former represent the state-of-the-art software for solving mathematical programming problems, we recognize that many schools and institutions may only have access to free or open solutions. Hence, we include both types of solvers in the experiments. More specifically, we selected two non-commercial (SCIP and CBC) and one commercial solver (Gurobi). All of the solvers were configured to have a time limit of 15 min per problem and to use a single thread.

We employed the 120 problem instances described in the previous subsection in our experimental setting. Each problem instance was solved with three configurations for α and β: $(\alpha = 0.5, \beta = 0.5)$, $(\alpha = 1, \beta = 0)$, and $(\alpha = 0, \beta = 1)$. The first configuration represents a balanced scenario where the same importance is given to the students' preferences and the companies' preferences for the allocation. The second and third configuration represent scenarios where only the students' preferences are considered, and only the companies' preferences are considered, respectively. These different configurations are employed to assess the conflict between the different objectives of the model. Each problem instance and configuration was solved 30 different times to capture stochastic variations in the computation time. Thus, this gives a total of $120 \times 3 \times 3 \times 10 = 10,800$ problem resolutions. We collected information about the execution time and the value of the objective function for both the students and companies. The experiments were run on 2.2 GHz processors 16 GB of RAM.

4.3 Results

We aggregated all of the repetitions into mean values for the same problem instance, configuration, and solver. Then, with the resulting dataset we conducted an ANOVA analysis using the execution time as response variable and the rest of the variables (i.e., number of students, surplus, number of primary skills, configurations for α and β, and the solver) as factors. The main goal of this analysis is understanding how the different variations of the problem and the solver type may affect the execution time of the model and, thus, its resolution in practice.

Table 1. ANOVA type III ($\alpha = 0.05$) analysis taking the execution time as a response variable. A bold font is employed to highlight significant p-values.

	Df	Sum squares	Mean Squared	F value	p-value
A: Configuration for alpha and beta	2	1.68E+11	8.43E+10	1.68	0.1869
B: Primary competences	1	5.91E+9	5.91E+9	0.12	0.7315
C: Solver type	2	3.12E+13	1.56E+13	311.42	**0.00**
D: Number of students	3	7.58E+13	2.52E+13	503.61	**0.00**
E: Surplus	2	6.69E+12	3.34E+12	66.69	**0.00**
Interaction AB	2	7.44E+10	3.72E+10	0.74	0.4769
Interaction AC	4	1.50E+11	3.76E+10	0.75	0.5579
Interaction AD	6	3.28E+11	5.48E+10	1.09	0.3658
Interaction AE	4	8.11E+10	2.02E+10	0.40	0.8059
Interaction BC	2	4.21E+10	2.79E+10	0.42	0.6576
Interaction BD	3	8.37E+10	1.74E+10	0.56	0.6442
Interaction BE	2	3.48E+10	5.19E+12	0.35	0.7006
Interaction CD	6	3.11E+13	4.29E+12	103.48	**0.00**
Interaction CE	4	1.71E+13	8.00E+11	85.50	**0.00**
Interaction DE	6	4.80E+12	5.02E+10	15.94	**0.00**
Residuals	1030	5.17E+13	3,97E+11		

The ANOVA table can be found in Table 1. The table contains the study of the effect of the aforementioned factors on the mean execution time, as well as the effect of all possible interactions of the second degree. In addition to this, we have included plots with the means and the Tukey intervals ($\alpha = 0.05$) for those factors that were found to have a significant effect on the mean execution time (Fig. 1). Finally, we include interaction plots for those second-order interactions that were found significant. From the main factors, we found the number of students, the surplus, and the solver type to have a significant effect on the average execution time. As observed in Fig. 1, Gurobi is the fastest solver.

Let us start by analyzing the effect of the number of students. As we expected, Table 1 suggests that the number of students has a significant effect on the mean execution time. This is natural, as it controls the number of decision variables and the number of constraints of the model. Hence, as one could expect, the more decision variables and constraints the more execution time it takes to solve the problem, as they are related to the size of the problem. In Fig. 1 one can observe the increasing tendency of the mean execution time and the number of students. It can be highlighted that the mean execution time grows in a possibly exponential way. As a consequence, one conclusion that can be drawn from these experiments is the necessity for alternative methods to solve the problem

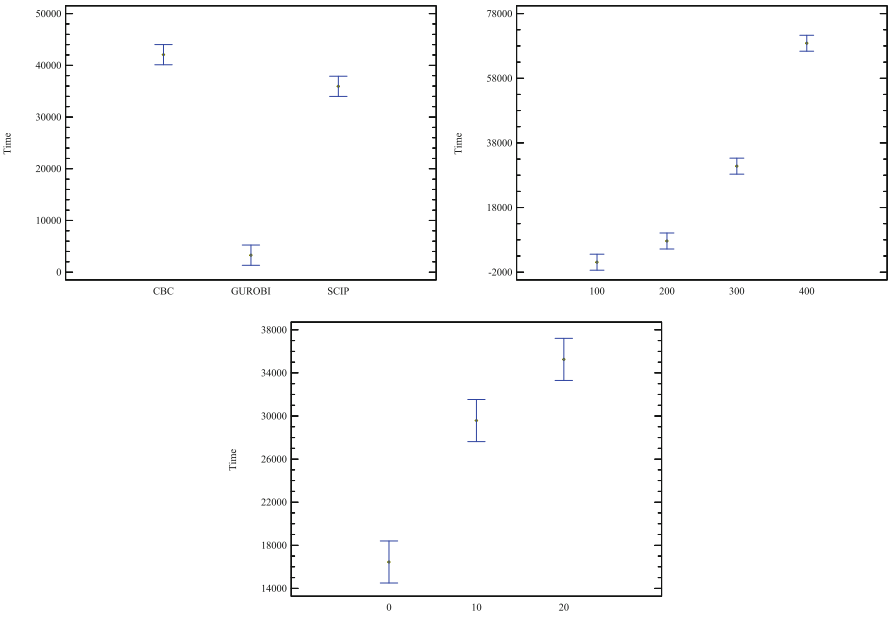


Fig. 1. Means Plot and Tukey HSD intervals (95% confidence) for the factor type of solver (left), the number of students (right), and the surplus (bottom)

when one faces larger problem instances. The interaction between the number of students and the solver type was found significant (see Fig. 2, suggesting a different growth for the execution time when increasing the number of students attending to the solver type. Effectively, Gurobi has a slower growth compared to CBC and SCIP, being the fastest of the three solvers and showing consistent behavior across problem instances. Despite showing a good performance, not all schools and educational institutions may have access to commercial solvers. Hence, they may need to employ non-commercial solvers, which have a faster growth for the mean execution time and whose mean execution times may result in unacceptable for larger problem instances.

Some similar conclusions can be drawn by analyzing the effect of the surplus on the mean execution time (Fig. 1 and Table 1). The larger the surplus, the longer the mean execution time. This is also natural, as more placements mean more decision variables and, therefore, a larger problem size. What is interesting is the fact that not all of the solvers show the same behavior when facing problems with more placement opportunities than students, as shown in the interaction plot between solver and surplus (see Fig. 2). CBC shows very competitive execution times when the number of students is equal to the number of placements, being close to the times achieved by Gurobi, the fastest among the three solvers. However, this performance quickly deteriorates when the number of available placements arises, becoming the worst option among the three solvers. SCIP shows a more stable behavior as well as Gurobi. We can attribute

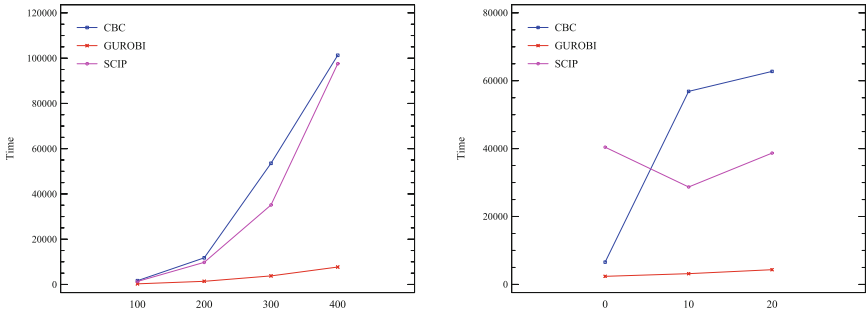


Fig. 2. Interaction plot of the number of students and surplus and the solver

Gurobi's better performance to the more sophisticated heuristic implementations it integrates, as well as other optimizations. This performance depends on the type of problem, but in our specific problem, the performance of the commercial software is clearly superior.

5 Conclusions

In this paper, we have introduced a multi-objective integer linear programming model for the SIPA problem, which is focused on matching students with industry placements. This proposal would be particularly suitable for environments that require the assignment of students to practical work, such as Vocational Educational Training or high school. This model incorporates the student and company preferences with a multi-objective approach. Preferences are defined through the competences associated with the current development level of students, their competence development preferences within the placement, and the competence requirements specified by each placement. In addition, the model also incorporates the distance between the student and the placement site, as well as interdependencies between placements.

We formally defined the SIPA problem and a model to address it. Then, we conducted an experimental comparison of various commercial and non-commercial solvers to analyze the scalability and performance of the model. To do this, we created different instances using real student and company data. According to the results, the non-commercial solvers do not scale as well as the Gurobi commercial solver. However, it is important to consider that many schools and educational institutions do not have access to commercial solvers and must rely on non-commercial software. Furthermore, while Gurobi shows fairly scalable and efficient behavior, it is important to note that its growth is exponential, which can be a drawback when dealing with larger instances than those used in these experiments. Therefore, as part of future work, we are working on defining a heuristic or metaheuristic method to solve instances of larger problems within a reasonable timeframe. It is worth noting that lecturers are another

relevant actor to consider in future versions of this model, as they need to supervise students. Hence, as part of future work, we aim to develop a model that also takes into account lecturer preferences for supervising student placements and, consequently, making visits to the companies. Finally, aim at conducting user evaluations to measure the effectiveness and pedagogical relevance of our approach.

Acknowledgements. This work is partially funded by the Spanish Ministry of Science and Innovation under the projects PID2021-124975OB-I00 and PID2021-123673OB-C31, as well as the project TED2021-131295B-C32 from the State Research Agency and the project DIGITAL2022 CLOUDAI02/S8760000 from the European Commission.

References

1. Ali, A.Y.: Greedy algorithm for solving student allocation problem in internship program: a case study. Int. J. Res. Ind. Eng. **10**(2), 155–164 (2021)
2. Anwar, A.A., Bahaj, A.: Student project allocation using integer programming. IEEE Trans. Educ. **46**(3), 359–367 (2003)
3. Binder, J.F., Baguley, T., Crook, C., Miller, F.: The academic value of internships: benefits across disciplines and student backgrounds. Contemp. Educ. Psychol. **41**, 73–82 (2015)
4. Chiarandini, M., Fagerberg, R., Gualandi, S.: Handling preferences in student-project allocation. Ann. Oper. Res. **275**(1), 39–78 (2019)
5. Harper, P., de Senna, V., Vieira, I., Shahani, A.: A genetic algorithm for the project assignment problem. Comput. Oper. Res. **32**(5), 1255–1265 (2005)
6. Manlove, D., Milne, D., Olaosebikan, S.: An integer programming approach to the student-project allocation problem with preferences over projects. In: International Symposium on Combinatorial Optimization, pp. 313–325. Springer (2018)
7. Nand, R., Sharma, A., Reddy, K.: Skill-based group allocation of students for project-based learning courses using genetic algorithm. In: IEEE International Conference on Teaching, Assessment, and Learning for Engineering, pp. 394–400. IEEE (2018)
8. Narayanan, V.K., Olk, P.M., Fukami, C.V.: Determinants of internship effectiveness: an exploratory model. Acad. Manag. Learn. Educ. **9**(1), 61–80 (2010)
9. Pan, L., Chu, S., Han, G., Huang, J.Z.: Multi-criteria student project allocation: a case study of goal programming formulation with DSS implementation. In: The 8th International Symposium on Operations Research and Its Applications, pp. 75–82 (2009)
10. Salah, M., Sanchez-Anguix, V., Alberola, J.M., Villa, F.: An integer linear programming model based on competences of student-industry placement allocation. In: The 39th ACM/SIGAPP Symposium on Applied Computing, pp. 104–105 (2024)
11. Salami, H.O., Mamman, E.Y.: A genetic algorithm for allocating project supervisors to students. Int. J. Intell. Syst. Appl. **8**(10), 51 (2016)
12. Sanchez-Anguix, V., Alberola, J.M., Del Val, E., Palomares, A., Teruel, M.D.: Comparing computational algorithms for team formation in the classroom: a classroom experience. Appl. Intell. **53**(20), 23883–23904 (2023)

13. Sanchez-Anguix, V., Chalumuri, R., Aydoğan, R., Julian, V.: A near pareto optimal approach to student-supervisor allocation with two sided preferences and workload balance. Appl. Soft Comput. **76**, 1–15 (2019)
14. Şimşek, A.B.: A decision support tool for the student-supervisor allocation problem of postgraduate programs. Expert Syst. Appl. **190**, 116068 (2022)
15. Srinivasan, D., Rachmawati, L.: Efficient fuzzy evolutionary algorithm-based approach for solving the student project allocation problem. IEEE Trans. Educ. **51**(4), 439–447 (2008)
16. Thiruvady, D., Morgan, K., Bedingfield, S., Nazari, A.: Allocating students to industry placements using integer programming and ant colony optimisation. Algorithms **14**(8), 219 (2021)